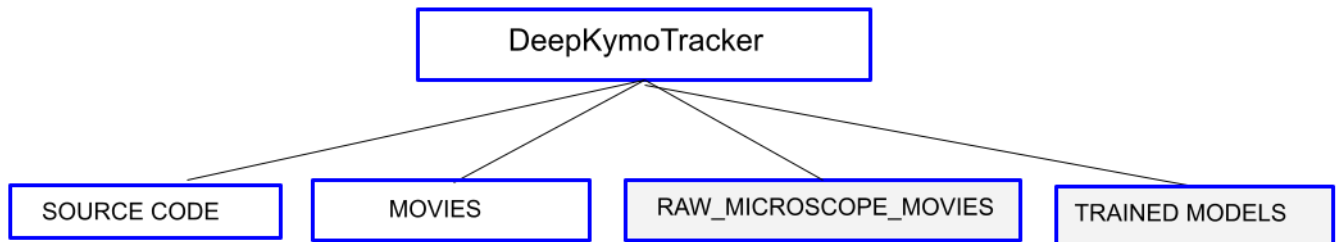


3. How to set up DeepKymoTracker folder (User guide)

Here is the folder tree structure for DeepKymoTracker you are supposed to have in place before starting execution:



The Complete Setup Process For The User

1. Create DeepKymoTracker folder.

Create an empty folder named **DeepKymoTracker** anywhere on your computer.

2. Download source code from GitHub.

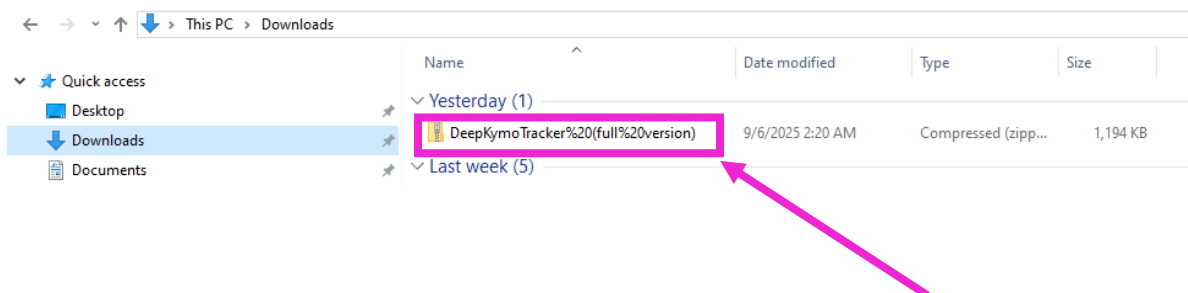
Copy URL below:

https://github.com/khelina/T-cell-lineages-tracking/tree/main/DeepKymoTracker_full

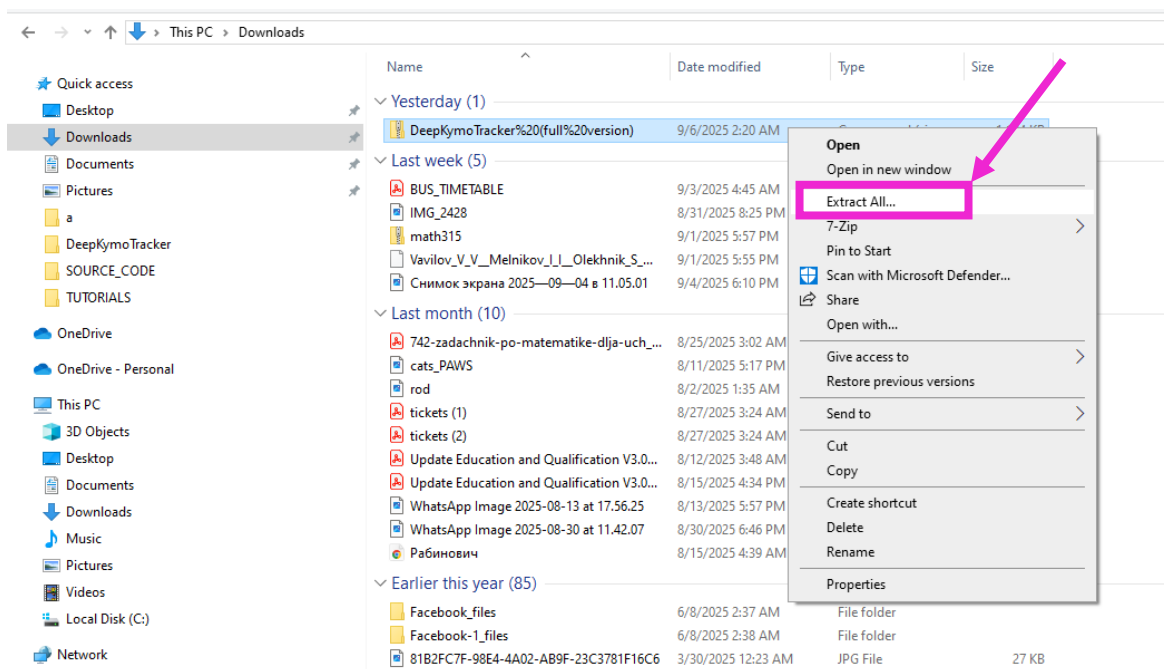
then go [DownGit](#) , enter the URL and hit Download button:

Go to [DownGit](#) → Enter Your URL → Download

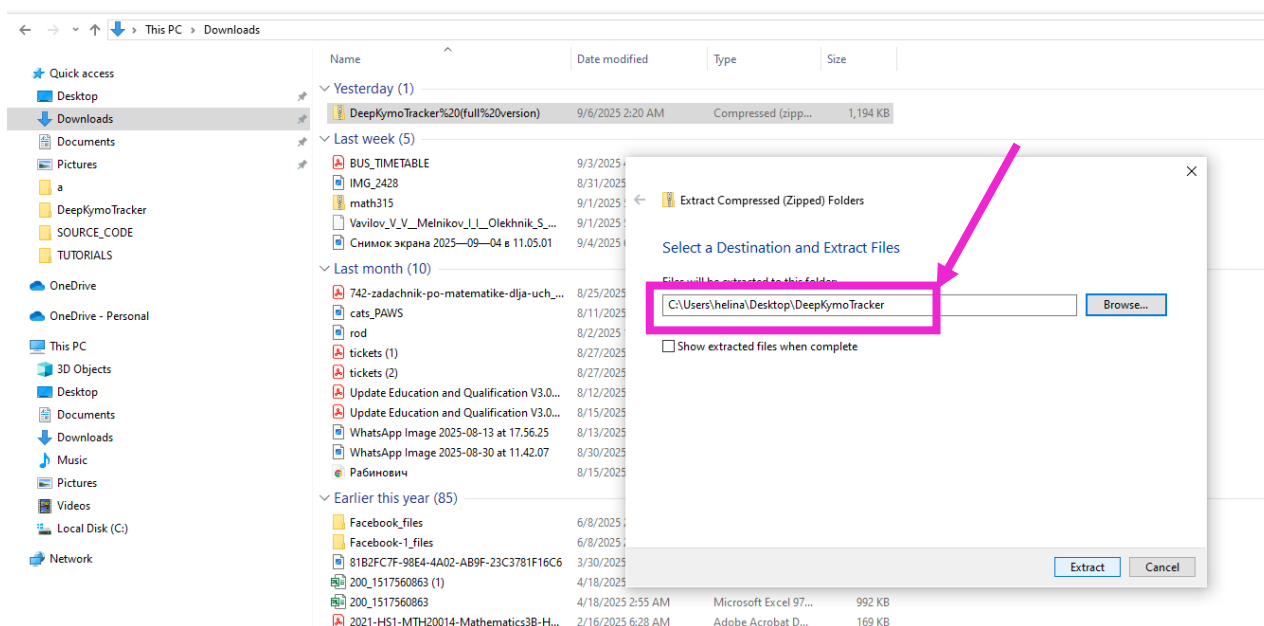
The zip file with the name `DeepKymoTracker%20(full%20version).zip` will appear in your Downloads folder.



To unzip it, right-click on it and choose Extract All from the dropdown menu:



Using Browse button, choose the directory to your DeepKymoTracker folder you just created and hit Extract:



The folder will land in your DeepKymoTracker folder. Change the folder name to become SOURCE CODE:

DeepKymoTracker%20(full%20version) → **SOURCE CODE**

3. Run GUI_launch.py.

Inside SOURCE_CODE, find file GUI_launch.py and launch it. It will automatically create subfolders TRAINED MODELS, MOVIES and RAW_MICROSCOPE_MOVIES, and will give you instructions how to fill these folders.

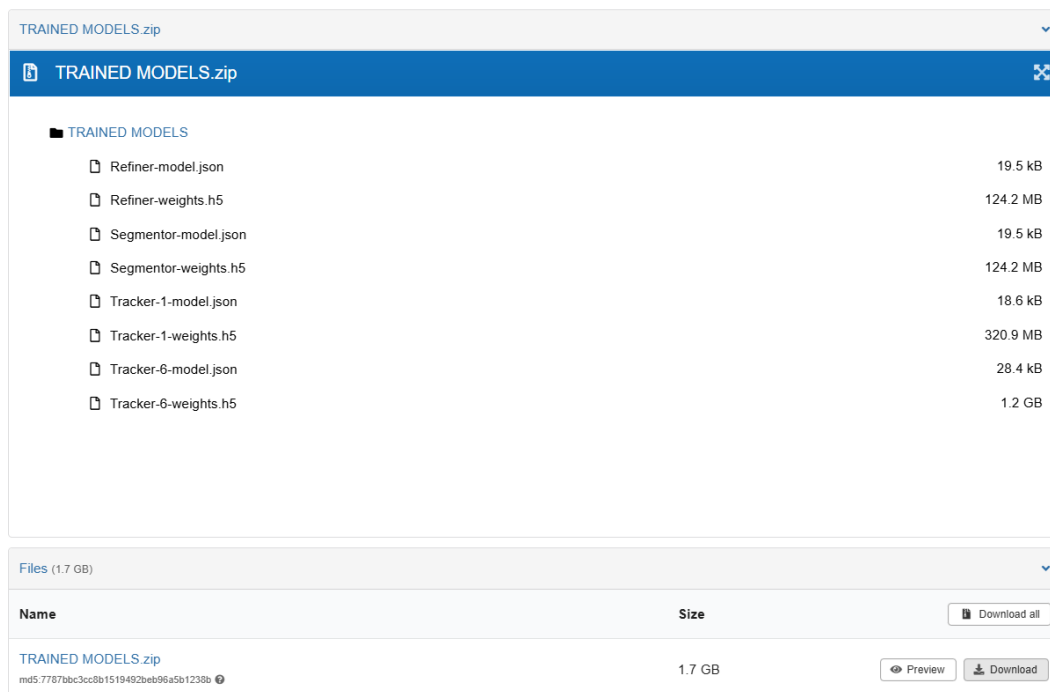
4. How to fill TRAINED MODELS folder:

Download models from Zenodo.

To get to Zenodo repository, click on the link below:

<https://doi.org/10.5281/zenodo.11540886>

The contents of the repository will appear. Hit Download button at the bottom of the screen:



The screenshot displays the Zenodo interface for a repository. At the top, a header bar shows 'TRAINED MODELS.zip' with a dropdown arrow. Below this, a blue banner also displays 'TRAINED MODELS.zip' with a file icon and a zoom icon. The main content area lists the files within the zip, organized under a 'TRAINED MODELS' folder icon. The files and their sizes are:

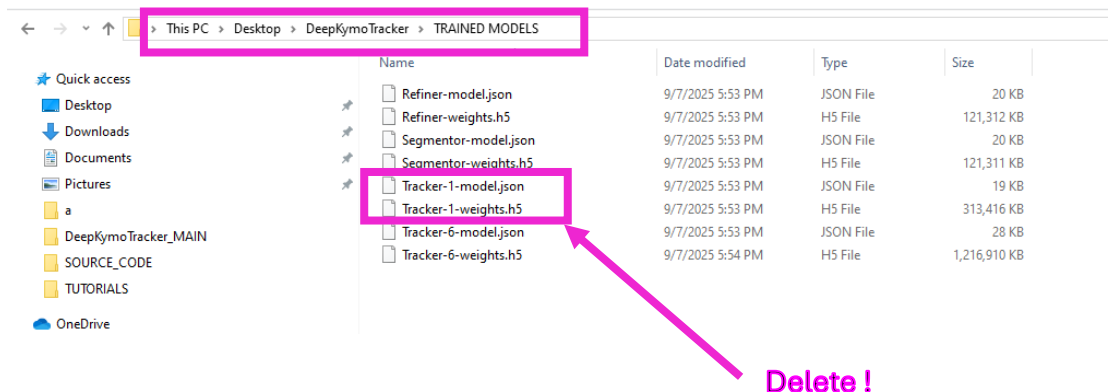
File Name	Size
Refiner-model.json	19.5 kB
Refiner-weights.h5	124.2 MB
Segmentor-model.json	19.5 kB
Segmentor-weights.h5	124.2 MB
Tracker-1-model.json	18.6 kB
Tracker-1-weights.h5	320.9 MB
Tracker-6-model.json	28.4 kB
Tracker-6-weights.h5	1.2 GB

At the bottom of the page, a 'Files (1.7 GB)' section provides a summary table:

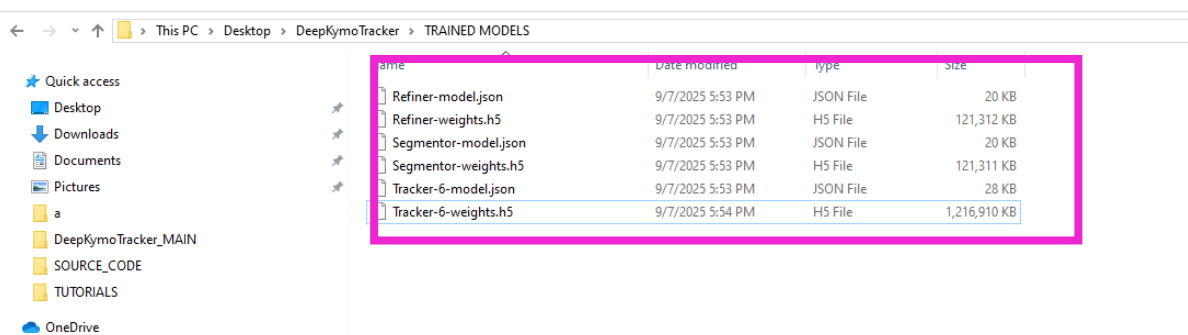
Name	Size	Download all
TRAINED MODELS.zip md5:7787bbc3cc8b1519492beb96a5b1238b	1.7 GB	Download all

Below the summary table, there are buttons for 'Preview' and 'Download'.

– they are from the old version of DeepKymoTracker and are not utilised any longer:



Check the result:



5. How to fill MOVIES and RAW_MICROSCOPE_MOVIES folders

- **RAW_MICROSCOPE_MOVIES** folder is where you need to place folders containing multiple movies taken by spinning disc or epi. Examples of folder names: 20181218 DN2B Ep, 20190206 DN2B EpCAM.

Images file names: they need to contain

- **_w2FITC_** or **_w3Multi600_** or **_w4Alexa-633_** (green fluorescent channel),
- **_w1BF_** (brightfield channel),
- **_w3TRITC_** (red fluorescent channel).

If you need to track and segment the movies inside such folders, you will have to start with Step-1: EXTRACT MOVIES of DeepKymoTracker pipeline.

- If you already have an extracted movie given in green fluorescent and brightfield (and optionally, red fluorescent) channels:
 - The input cell movie needs to have images in 2 channels – green fluorescent and brightfield (and optionally, red fluorescent) channels.
 - Frame names format: should end with **_t00002_ch00.tif** (for the green fluorescent channel) , **_t00002_ch02.tif** (for the bright field channel), **_t00002_ch01.tif** (for the red fluorescent channel). In this example, t00002 is the number of the frame.
 - Image data type: uint8

You need to create a separate subfolder for that movie inside **MOVIES** folder. For example, if you want to give name TEST to your movie, you need:

1. Create a folder named TEST inside MOVIES folder;
2. Place the folder with your movie inside TEST folder and rename it as RAW_MOVIE_TEST.

